WORKSHEET

Binomial probability – Mean and standard deviation

1 At a rural high school, 28% of students travel more than 30 km each day to school. In a class of 25 students:

a what is the probability that five students travel more than 30 km each day to attend school?

b what is the probability that fewer than three students travel more than 30 km each day to attend school?

c what is the expected number of students travelling more than 30 km?

d what is the standard deviation?

2 A particular vaccine protects 85% of people from contracting a disease. In a group of 12 immunised people:

a what is the probability that none of them contract the disease?

b what is the probability that five people contract the disease?

c what is the expected number of people protected from the disease?

d what is the standard deviation?

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3 Sixty-six per cent of university graduates earn more than $30 000 in their first year out. A survey is conducted on 100 recent university graduates.

a What is the probability that 50 of them earned over $30 000 in their first year out?

b What is the probability that 35 of them earned no more than $30 000 in their first year out?

c What is the expected number of graduates earning over $30 000 in their first year out?

d What is the standard deviation?

4 Approximately 27% of people have blue eyes. In a group of 12 people:

a what is the probability that five people have blue eyes?

b what is the probability that fewer than two people have blue eyes?

c what is the probability that more than nine people have brown, grey or green eyes?

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d what is the expected number of people in the group with blue eyes?

e what is the standard deviation?

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Answers

1 **a** ≈ 0.1282

b ≈ 0.0152

c 7

d ≈ 2.2450

2 **a** ≈ 0.1422

b ≈ 0.0193

c 10.2

d ≈ 1.2369

3 **a** ≈ 0.0004

b ≈ 0.0816

c 66

d ≈ 4.7371

4 **a** ≈ 0.1255

b ≈ 0.1245

c ≈ 0.3313

d 3.24

e ≈ 1.5379

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